

VisionAssist AI

Complete Independence for the Visually Impaired

₹1,399

Total Cost

10+

AI Features

100%

Open Source

4 NEW

Features

THE PROBLEM

2.2 Billion people live with vision impairment — WHO 2024

Cannot Navigate Safely

Obstacles & potholes cause daily injuries. No affordable AI navigation.

Cannot Read Documents

Aadhaar, bills, books — all inaccessible without constant human help.

Cannot Identify Food

Can't identify meals, quantities, or plate contents independently.

Cannot Cross Roads Safely

Traffic light detection requires sighted help — a daily danger.

Cannot Identify Faces

Recognizing family requires touch or voices — limiting social life.

Cannot Handle Currency

Counting notes by feel is slow, error-prone, and embarrassing.

OUR SOLUTION

₹1,399

vs ₹3,00,000 OrCam Smart Glasses

VisionAssist AI combines an Arduino UNO with dual sonar sensors and a smartphone to deliver 10+ AI features — object detection, food identification, stair navigation, traffic light reading, face recognition, currency detection, document scanning, and emergency SOS — all controlled by the 'My Eye' voice assistant.

- 01 Universal Accessibility —** ₹3L smart glass capabilities at ₹1,399 via open-source AI
- 02 Complete AI Feature Suite —** 10+ features — food, stairs, traffic, face, currency, SOS, navigation
- 03 Real-Time Safety —** Dual-sonar + AI vision detects head & foot obstacles simultaneously
- 04 Voice-First 'My Eye' —** Zero touch required — natural language, instant spoken feedback
- 05 Scalable Open Source —** ₹0 software cost · scales to ₹800/unit · NGO & govt deployment ready

SYSTEM ARCHITECTURE

HARDWARE

Arduino UNO

HC-SR04 Sonar (head level)

HC-SR04 Sonar (31° foot level)

Smartphone Camera

Custom PVC Case

USB/Bluetooth → Flask

Voice Wake
'My Eye'

Intent
Parsing

Camera
Capture

Groq AI
Vision

TTS
Output

NORMAL MODE

- ▶ 4-6 frames over 3-4 seconds
- ▶ Best frames → Groq Llama-4 Vision
- ▶ Detailed spoken description
- ▶ Best for: Food · Stairs · Documents

LIVE MODE

- ▶ Continuous 3-frame capture cycle
- ▶ TTS completes before next call
- ▶ No overlapping speech
- ▶ Best for: Traffic · Stair monitoring

TECHNOLOGY STACK

AI:

Groq Llama-4 Scout · YOLOv8n ·
DeepFace · SIFT

Backend:

Python Flask · PIL preprocessing ·
YOLO+HSV

RAG:

n8n Automation · Pinecone Vector DB ·
Doc Q&A

Voice:

WebSpeech API · TTS queue · Wake
word 'My Eye'

4 NEW AI FEATURES

Built on Groq Llama-4 Scout Vision

Food Detector

- Identifies all food items, quantities & meal type
- Normal: 4 frames / 3s · Live: continuous scan
- Works for breakfast, lunch, dinner, snacks

Stair Detector

- Counts steps (exact or estimated) ± 1 step accuracy
- States direction: UP or DOWN
- Reports handrail position, hazards & lighting

Traffic Light Detector

- Groq vision + parallel HSV color analysis
- YOLO detects traffic light ROI for precision
- Live: only speaks when light state changes

360° Object Finder

- User asks any natural language question
- Captures 5 frames of scene for analysis
- AI locates object: direction, distance, landmarks

VALIDATED OUTCOMES

Tested with real visually impaired users

DEPLOYED

Real-Time Obstacle Avoidance

Dual-sonar detects obstacles at head & foot level simultaneously — tested across corridors, streets & staircases

LIVE TESTED

Traffic-Safe Road Crossing

Groq + YOLO + HSV triple-layer analysis correctly identifies RED / GREEN in complex real-world intersections

USER VERIFIED

Full Document Accessibility

RAG pipeline reads Aadhaar, utility bills & books aloud — zero human assistance required in all test sessions

FEATURE · WHAT IT DELIVERS · VALIDATED OUTCOME

Food Detector

Identifies meal, items & quantities aloud

✓ Works across thali, packed meals & beverages

Stair Detector

Step count, direction UP/DOWN, handrail & hazards

✓ Tested on 6 staircase types — indoor & outdoor

360° Object Finder

Locates any named object by direction & distance

✓ Finds keys, bags, chairs in 5-frame scene scan

Currency Detection

Reads all Indian denominations via SIFT matching

✓ Works in wallet, hand, low-light conditions

Face Recognition

Instant spoken ID of registered contacts

✓ DeepFace + FaceNet — tested at 1–3m range

IMPACT & SCALABILITY



Complete Independence

10+ daily tasks freed
Walks, reads, cooks, crosses roads —
unassisted



200× More Affordable

₹1,399 vs ₹3,00,000 OrCam
Scales to ₹800/unit at manufacturing

5M

National Scale Target

5M+ visually impaired Indians
Govt ADIP scheme eligible · NGO-ready

SCALE-UP ROADMAP

Phase 1 NOW

Arduino + Smartphone
₹1,399 prototype

Phase 2 3D PRINT

Custom 3D-printed case
₹1,799 polished unit

Phase 3 SCALE

Bulk manufacturing
₹800/unit with NGO

Phase 4 ECOSYSTEM

Android app + edge AI
Offline & cloud hybrid

Target: 5M+ in India · 75M globally (NSSO + WHO)

CHALLENGES & FUTURE ROADMAP



CHALLENGES FACED & SOLUTIONS

Groq API rate limits in live mode

→ Solved with TTS-wait pattern

WebM video codec failing on Windows

→ Fixed by sending base64 frames JSON

OpenCV unavailable on some systems

→ YOLO+HSV made optional with graceful fallback

Traffic light false positives in busy scenes

→ YOLO+HSV consensus layer added

Food detection w/ overlapping items

→ Multi-frame averaging mitigates errors

Limited Indian food training data

→ Using Groq's broad vision knowledge

Low-light stair detection

→ Frame contrast enhancement preprocessing



FUTURE ENHANCEMENTS



Dedicated Android App

Offline mode — core features without internet



On-Device Edge AI

TFLite/ONNX — zero-latency, zero data cost



Bus & Metro Detection

Camera reads bus numbers & route info aloud



Bone-Conduction Audio

Hands-free, discreet earphone integration



Obstacle Map Memory

Learns frequent routes, pre-warns on hazards



Multi-Language Support

Marathi, Hindi, Tamil voice responses



Govt Integration

ADIP certification · NIVH partnership roadmap



NOT JUST AN INNOVATION

A REVOLUTION

- ✓ ₹1,399 · Replaces ₹3,00,000 smart glasses
- ✓ 10+ AI features: Food · Stairs · Traffic · Face · Currency · Navigation · SOS
- ✓ Dual-sonar: simultaneous head & foot obstacle detection
- ✓ Groq Llama-4 Scout · YOLOv8 · DeepFace · SIFT · 100% open source
- ✓ NGO · Government · Hospital deployment ready · Scales to ₹800/unit